PAPER ID-311224

Roll No: | | | | | | | | | | | | | |

BTECH-MINOR
(SEM V) THEORY EXAMINATION 2024-25
INTRODUCTION TO SECURITY OF CYBERPHYSICAL SYSTEMS

TIME: 3 HRS **M.MARKS: 100**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt *all* questions in brief. **2 x 10 = 20**

Q no.	Question	CO	Level
a.	Explain the relationship between security and privacy in information systems.	1	K1, K2
b.	How do organizations identify, assess, and mitigate risks in complex information systems?	1	K1, K2
c.	What are the potential risks of relying on AI-driven detection mechanisms?	2	K1, K2
d.	Discuss the challenges of implementing encryption in resource-constrained environments.	2	K1, K2
e.	How can organizations ensure privacy in smart home environments?	3	K2
f.	Discuss the concept of Bring Your Own Device (BYOD) in IoT security.	3	K2
g.	Explore the role of machine learning in enhancing SDN security.	4	K2, K3
h.	What are the advantages and challenges of SDN?	4	K2, K3
i.	Discuss the impact of machine learning on CPS security.	4	K3, K6
j.	How do case studies of CPS security breaches provide insights into improving system resilience?	4	K3, K6

SECTION B

2. Attempt any *three* of the following: **10 x 3 = 20**

Q no.	Question	CO	Level
a.	Evaluate the impact of data breaches on privacy and security. Discuss the measures organizations can take to recover from such incidents and restore user trust.	1	K1, K2
b.	Explain the relationship between information theory and intrusion detection. How does Shannon's entropy help in detecting anomalies?	2	K1, K2
c.	Analyze the unique security challenges of IoT devices. How do these challenges differ from traditional IT systems?	3	K2
d.	Discuss the potential attack vectors in SDN architecture. How can these vectors be identified and addressed?	4	K2, K3
e.	Explain the key components of a Cyber-Physical System. How do these components interact to ensure system functionality?	4	K2, K3

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SECTION C

3. Attempt any *one* part of the following: 10 x 1 = 10

Q no.	Question	CO	Level
a.	Discuss the significance of end-user education in preventing security breaches. What strategies can organizations use to improve user awareness about privacy and security?	1	K1, K2
b.	Analyze the trade-offs between usability and security in information systems. How can designers strike a balance to ensure effective user experiences without compromising security?	1	K1, K2

4. Attempt any *one* part of the following: 10 x 1 = 10

Q no.	Question	CO	Level
a.	Describe the Diffie-Hellman key exchange mechanism. How does it ensure secure communication between parties over an untrusted network?	2	K1, K2
b.	Compare and contrast symmetric and asymmetric encryption algorithms. How are they applied in ensuring confidentiality in real-world systems?	2	K1, K2

5. Attempt any *one* part of the following: 10 x 1 = 10

Q no.	Question	CO	Level
a.	How does edge computing influence IoT security? Discuss its advantages and challenges in securing IoT systems.	3	K2
b.	Discuss the potential impact of AI on IoT security. How can AI be leveraged to predict and prevent security threats?	3	K2

6. Attempt any *one* part of the following: 10 x 1 = 10

Q no.	Question	CO	Level
a.	Explain the architecture of Software-Defined Networks. How does it differ from traditional networking paradigms?	4	K2, K3
b.	Discuss the concept of privacy leakage in SDN environments. How can organizations mitigate these risks?	4	K2, K3

7. Attempt any *one* part of the following: 10 x 1 = 10

Q no.	Question	CO	Level
a.	Discuss the future of CPS security. How will emerging technologies influence the design and protection of CPS?	4	K2, K3
b.	Analyze the security vulnerabilities of intelligent CPS. What measures can be taken to secure CPS deployments?	4	K2, K3

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